

MODULE 1

A. Compounding Technique/ Future Value Technique

Future value of a single flow (Lumpsum) :

$$FV = P(1+r)^n$$

FV= Future Value

P= Principal

r = Interest rate per period

n = number of periods/ time period

1. Calculate the future value of Rs 50,000 at the end of 3 years at 12% pa rate of interest
2. Calculate the future value of Rs 20,000 invested now for 5 years at a time preference rate of 8% pa
3. Mr Krishna deposits Rs 20,000 for 3 years at 10% interest. What is the compounded value of his deposit?
4. Mr Hemanth deposits Rs 30,000 for 3 years at 10% pa interest. Calculate the maturity value of the deposit (FV), if the interest is compounded semi-annually (half-yearly).
5. Mr Ramesh invests Rs 1,00,000 in a bank at 10% pa for 5 years. Calculate the maturity value if interest is compounded annually. Will he get more if interest is compounded half-yearly?

Future value of multiple flows or a series of flows

a) When investments occur at the end of each period:

$$FV = P_1 (1+r)^{n-1} + P_2 (1+r)^{n-2} + P_3 (1+r)^{n-3} + \dots + P_n$$

Where,

FV= Future Value

P= Principal

r = Interest rate per period

n = number of periods/ time period

6. Determine the FV at the end of 5 years pf the following series of payment at 8% rate of interest
At the end of 1st year Rs 1,000
At the end of 2nd year Rs 2,000
At the end of 3rd year Rs 3,000
At the end of 4th year Rs 4,000
At the end of 5th year Rs 5,000

7. Calculate the compound value if the following payments are made at the end of each year:

Year	1	2	3	4
Payment	4,000	5,000	6,000	7,000
Rate of interest is 12% pa				

8. Calculate the compound value if the following payments are made at the end of each year:

Year	1	2	3	4	5
Payment	2,000	3,000	4,000	5,000	6,000
Rate of interest is 10% pa					

b) When investments occur at the beginning of each period:

$$FV = P_1 (1+r)^{n-1} + P_2 (1+r)^{n-2} + P_3 (1+r)^{n-3} + \dots + P_n (1+r)$$

Where,

FV= Future Value

P= Principal

r = Interest rate per period

n = number of periods/ time period

9. We invest Rs 1,000 at the beginning of 1st year Rs 2,000 at the beginning of 2nd year and Rs 3,000 at the beginning of 3rd year. How much will these flows accumulate at the end of 3rd year at a rate of interest of 12% pa.
10. X deposits Rs 5,000 at the beginning of 1st year, Rs 10,000 at the beginning of 2nd year, Rs 15,000 at the beginning of 4th year and Rs 25,000 at the beginning of 5th year. The deposits earn an interest at 8% pa. Determine how much money he will have at the end of 5 years?

Future value of an annuity

- Mr X deposits Rs 10,000 at the end of every year for 4 years and the deposit earns a compound interest at 10% pa. Determine how much money he will have at the end of 4 years.
- Mr A deposits Rs 5,000 at the end of every year and the deposit earns compound interest at 8% paa. Determine how much he will have at the end of 5 years?

B. Discounting Technique/ Present Value Technique

Present value of a single flow

$$PV = FV / (1+r)^n$$

Where,

PV = Present value

A = Amount at the end of period 'n'

r = Interest rate

n = n years

16. Mr X is to receive from Indira Vikas Patra Rs 20,000 after 5 years from now. His time preference for money is 10% pa. Calculate its present value.
17. Calculate the present value of Rs 10,00,000 to be received after 5 years from now assuming 6% time preference for money.

Present value of multiple flows or a series of flows

18. Calculate the present value of the following cash flows assuming a discount rate of 8% pa

Year	Cash Flows
1	20,000
2	40,000
3	20,000
4	10,000

19. Calculate the present value of the following cash flows

Year	Cash Flows
1	20,000
2	40,000
3	20,000
4	10,000

Note: Interest rate at 10% pa

Present value of an annuity

20. Determine the PV of an annuity of Rs 1,00,000 receivable for 5 years at an effective rate of interest of 12% pa.
21. Determine the PV of an annuity of Rs 20,000 receivable for 6 years at an effective interest rate of 10% pa.

